Cross validation

→ It's a technique used in ML to evaluate how well a model will generalize to new unseen data.

→ It prevents overfitting by averaging the model performance on multiple subsets of data.

→ By splitting data into training and testing sets multiple times, cross validation provides more robust estimate of model accuracy than a single train test split.

100 samples

80% training	20% testing

Random State

0 , 42 , 101

↑ Sample data

! Output

1> Loocv (Leave one out cross validation) :-

100 samples

test | train

Exit Random

Order Sampling

[0.2 , 0.3 , 0.1 , 0.5 , 0.6]

mean - accuracy ( )

2> k-fold (cross validation) :-

100 samples

test | train

1
2
3
4
5

k = 20

$\frac{100}{20} = 5$

mean - accuracy ( )